Does Chromatin Accessibility Explain Context-Dependent Knockdown Effects?

Research Proposal

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Summary

The same gene knockdown changes expression differently in different cell types, and it is not known how much of that difference chromatin accessibility explains once baseline expression is accounted for. This project measures it.

I use CRISPRi Perturb-seq screens that knock down the same set of roughly 2,000 essential genes in four human cell lines: K562, RPE1, Jurkat and HepG2. Two pairs carry the main test, K562-Jurkat and RPE1-HepG2, because the lines in each pair match on p53 status, tissue class and growth mode. The four mixed pairs are reported together as one contrast.

The work has four aims:

1. **Build the outcome.** Estimate how differently each gene responds to knockdowns in two lines, after removing sampling noise and the generic response all knockdowns share. Scale it against two screens of the same line.
2. **Test it correctly.** Use a randomization test whose null is built from the accessibility feature, so gene-gene correlation in the response cannot inflate significance.
3. **Prove the test works.** Show by simulation and by planting known signals into real data that the test controls false positives where standard tests do not.
4. **Confirm and extend.** Run the frozen test on the two matched pairs, contrast the mixed pairs, and check that the effect tracks how similar the accessibility differences are.

The headline result is one number with a confidence interval: the effect of accessibility difference on response difference in the matched pairs, beyond expression. It is reported against a same-line baseline, alongside how far a naive analysis overstates it.

Contribution in one sentence: a calibrated, replicated estimate of how much chromatin accessibility explains context-dependent knockdown effects, with evidence that standard gene-level tests get this wrong.

Background and gap

Cross-context divergence in knockdown effects has been measured but not explained. The TRADE preprint found the closest agreement for K562-Jurkat and HepG2-RPE1, with median correlations of 0.74 and 0.75. K562-RPE1 was lowest at 0.40. The authors note that the two close pairs share tissue class, p53 status and growth mode, which is why this design separates them.

The pieces of the method already exist, so this project does not claim them. It claims the measurement they make possible.

Prior work overview

Prior work	What it already does	What it leaves open
Replogle et al. 2022, genome-scale Perturb-seq	CRISPRi screens of essential genes in K562 and RPE1, plus a genome-wide K562 screen	Why the same knockdown differs between lines
TRADE (Nadig et al., Nature Genetics 2025)	Models estimation error in Perturb-seq effects; adds Jurkat and HepG2 screens of 2,393 essential genes each	What gene-level features explain the divergence it reports
Perturbation response decomposition (bioRxiv, July 2026)	Splits responses in four CRISPRi screens into global, perturbation-specific, cell-line-specific and interaction parts; ties the global part to proliferation and stress programs	Which gene-level features, chromatin included, explain the interaction part
CAMERA (Wu and Smyth 2012); Efron and Tibshirani 2007	Show that inter-gene correlation inflates gene-set tests	Application to cross-context divergence, where noise also varies by cell count
Conditional randomization test (Candès et al. 2018)	Gives a valid test by resampling the feature given covariates	A workable model of how accessibility is distributed across genes
Ahlmann-Eltze et al. 2025	Shows deep perturbation models do not yet beat simple linear baselines	Suggests careful statistics, not larger models, is the open route
ChromaPert (Ruan et al., arXiv, 6 October 2026)	Uses paired RNA and ATAC context in a deep model to predict responses across cell states in K562 multiome data	A calibrated, interpretable estimate across independent cell lines

Multiome Perturb-seq (Cell Systems)	Measures accessibility changes caused by knockdowns in RPE1	Whether pre-existing accessibility predicts the response

The question is hard for three statistical reasons:

- **Noise varies.** Divergence estimates are noisier for lowly expressed genes and for knockdowns with few cells.
- **Genes are not independent.** Co-regulated genes respond together, so thousands of genes carry far fewer independent observations.
- **Accessibility is confounded.** It tracks expression and copy number, and both also drive divergence.

There is also a biological reason to expect a subtle answer. A gene expressed in both lines usually has an open promoter in both, so promoter accessibility may vary too little to explain much. Distal elements such as enhancers are the more likely place for signal.

Novelty status: a search on the date above found no paper that reports this estimate. The July 2026 decomposition preprint comes closest and must be read in full first. A literature review with a mentor must confirm the gap before any claim of being first.

Hypotheses and predictions

Two hypotheses are primary, two are secondary, and one is explored through case studies. All are fixed in writing before any response difference is computed, each with a result that would count against it. Analyses are restricted to genes expressed above a set threshold in both lines of a pair.

Tier	Hypothesis	Supported if	If not, I report
Primary	**H1 Direction.** In the matched pairs, a gene more accessible in one line responds more strongly to knockdowns in that line.	The effect of signed accessibility difference on signed response difference is positive beyond expression, at 5% under the main test	That accessibility adds nothing detectable once p53, tissue and growth mode are held constant
Primary	**H3 Inflation.** Naive tests overstate the association.	Naive tests reject a true null well above 5% in simulation, and give a larger real-data effect than the main test	That naive analysis was adequate for this data type
Secondary	**H2 Location.** Distal regulatory accessibility explains more than promoter accessibility.	Distal feature has the larger effect, with non-overlapping intervals	Which feature carries the signal, or that neither does
Secondary	**H4 Similarity.** The effect of any accessibility difference on a pair's response difference rises with its similarity to that pair's own accessibility difference.	Effect increases with measured similarity across other pairs and decoys, and is near zero at low similarity	That the feature reflects generic gene properties, not context
Case studies	**H5 Mechanism.** For a transcription-factor knockdown, targets with binding sites open in only one line respond more in that line.	Described factor by factor, with a swap control; no pooled test	What individual factors show, without a general claim

Three controls decide whether any of this can be trusted:

- **Positive control.** Known p53 target genes should diverge between lines that differ in p53 function and not between lines that match. If the pipeline misses this, it is broken.
- **Negative control.** Decoy accessibility differences with near-zero similarity to the tested pair's own should explain nothing beyond chance.
- **Same-line control.** No accessibility difference should explain the response difference between two screens of K562.

A null result on H1 is still a finding if the test is validated. It would mean baseline expression already captures what accessibility knows about context.

Data

Everything needed is public, so the project requires no wet lab. Each item below still has to be checked against the actual files.

Resource	Use	Status
K562 and RPE1 essential-gene CRISPRi Perturb-seq (Replogle et al. 2022)	Pipeline pair; one line of each matched pair	Public. The TRADE preprint lists 2,057 and 2,393 knockdowns, with 2,053 shared across all four lines; count from files
K562 genome-wide Perturb-seq (Replogle et al. 2022)	Same-line baseline against the K562 essential screen	Public. Overlap of knockdowns and differences in protocol to confirm
Jurkat and HepG2 essential-gene Perturb-seq (Nadig et al. 2025)	Second line of each matched pair	Reported as 2,393 knockdowns each; confirm download

Resting and activated Jurkat screens, X-Atlas/Pisces	Same line in two states (extension)	Released on Hugging Face; about 10,900 knockdowns and 2.8 million cells per screen. Licence, compute and matching ATAC-seq unconfirmed
Non-targeting control cells from every screen	Noise null, baseline expression, reliability	Count cells per screen
K562 ATAC-seq, ENCODE (ENCSR868FGK)	Accessibility feature	Released; three replicates
hTERT-RPE1 ATAC-seq, 4DN (4DNES48BA2C8)	Accessibility feature	Released; wild type, two replicates. 4DN asks users to contact the lab before publishing
Jurkat and HepG2 ATAC-seq, ENCODE	Accessibility feature	Accessions to confirm
ATAC-seq from cell lines outside the screens, ENCODE	Decoy features, chosen by measured similarity	Choose lines processed by the same pipeline
Transcription-factor ChIP-seq in K562 and HepG2, ENCODE	Binding sites for H5 case studies	Count factors that are also in the knockdown library
Motif collection such as JASPAR	Binding sites where ChIP-seq is missing	Public
Copy number and TP53 status for each line	Confounds and positive control	Source for each line to confirm, RPE1 especially

Three features are computed per gene and per line, then differenced between lines:

- **Promoter accessibility:** normalized ATAC signal in a fixed window around the transcription start site.
- **Distal accessibility:** distance-weighted ATAC signal in peaks within a fixed window of the gene, promoter excluded.
- **Combined accessibility:** promoter plus distal, with weights fixed in advance. This is the feature for H1; the other two are compared in H2.

All are corrected for copy number, because ATAC signal scales with the number of DNA copies and K562 is heavily rearranged. Window sizes and the expression threshold are fixed before analysis.

Aim 1: Build the outcome and its baseline

Aim 1 builds the outcome on K562-RPE1 and freezes the pipeline there. That pair comes from one study and has a known p53 difference to check against. The matched pairs stay untouched until Aim 4.

1. **Effects.** For knockdown p, gene g and line k, estimate the log fold change against that line's own non-targeting control cells.
2. **Noise null.** Repeatedly split control cells into a pseudo-knockdown group, matched in cell count to knockdown p, and a pseudo-control group. The spread of these fake effects gives a noise variance for every real effect.
3. **Generic response.** For each gene and line, average the effect over all knockdowns and subtract it from every effect. This removes what every essential-gene knockdown does, such as stress and slowed growth.
4. **Response difference.** For each gene, average the noise-corrected squared specific effect over all knockdowns in each line, then take the difference between lines A and B.
5. **Reliability.** Split each line's cells into random halves and compute the response difference twice. The correlation r between halves is converted to full-data reliability, 2r/(1+r).
6. **Same-line baseline.** Run steps 1 to 5 on the two K562 screens over their shared knockdowns. Any response difference there is technical.
7. **Naive analysis.** Regress the response difference on expression difference and accessibility difference, treating genes as independent. Record the coefficient, partial R-squared and p-value.

Each effect first has its line's generic response removed:

$$\tilde{\beta}_{p,g,k} = \hat{\beta}_{p,g,k} - \frac{1}{P}\sum_{q=1}^{P}\hat{\beta}_{q,g,k}$$

The primary outcome for gene g is then:

$$M_g = \frac{1}{P}\sum_{p=1}^{P}\Bigl[\bigl(\tilde{\beta}_{p,g,A}^{2} - \tilde{s}_{p,g,A}^{2}\bigr) - \bigl(\tilde{\beta}_{p,g,B}^{2} - \tilde{s}_{p,g,B}^{2}\bigr)\Bigr]$$

Here $\tilde{s}^2$ is the noise variance after the same subtraction, found by passing the fake effects from step 2 through step 3. Removing it matters because a noisy estimate has a large square even when the true effect is zero. Without the correction, lowly expressed genes would look highly responsive.

The subtraction in step 3 splits the raw signal cleanly. A gene's mean squared effect equals the square of its average effect, the generic part, plus the spread around that average, the specific part. I report the generic share of the raw response difference for every pair. The July 2026 decomposition preprint tied this shared component to proliferation and stress programs, so a large share is expected.

A variant subtracts the average within preset knockdown groups, such as ribosome, proteasome and splicing, taken from outside annotation. It runs as a sensitivity analysis, because averages over small groups would absorb knockdown-specific signal.

M is directional: positive means the gene moves more in line A. It is tested against the signed accessibility difference, which makes H1 easy to falsify. An unsigned version, the noise-corrected mean of squared effect differences, is kept as a secondary outcome.

Covariates entered before accessibility are baseline expression in each line, expression difference, expression variance in control cells, copy-number difference and promoter CpG content. Knockdowns with weak on-target repression in either line are excluded by a preset cutoff.

Three checks gate everything after this aim:

- If p53 targets do not show a stronger response in RPE1 than in K562, the pipeline is wrong and gets fixed first.
- Every cross-line number is reported next to its same-line value: reproducible variance, generic share and naive effect.
- Every explained-variance figure is a share of reproducible variance, using the corrected reliability from step 5.

Aim 2: Test it correctly

The test holds the response data fixed and randomizes the accessibility feature. Its validity then depends only on how accessibility is distributed across genes, not on how gene responses correlate.

This corrects an error in the earlier draft. That draft grouped genes by co-expression of the response and then permuted the feature, which mixes two frameworks. When the feature is what gets randomized, dependence within the feature is what must be preserved.

Null hypothesis. Given the covariates C, the accessibility difference x is independent of the response difference M.

Procedure.

1. Compute the statistic T on the real data: the coefficient on x in a regression of M on C and x.
2. Draw a replacement feature from a model of x given C, and recompute T.
3. Repeat B times and take the p-value below.

$$p = \frac{1 + \#\{b : T^{*}_{b} \ge T\}}{1 + B}$$

Guarantee. If the replacement features are drawn from the true distribution of x given C, the p-value is exactly valid at any sample size. This holds whatever the correlation among genes in M, because M never changes. I will write out the proof, which rests on exchangeability of the real and replacement features under the null.

In practice that distribution is unknown. Block permutation is the main null, fixed in advance, and every primary result uses it. Stratified permutation and decoy features run as sensitivity checks, and any disagreement with the main null is reported.

Blocks are defined on the feature side: genes that are genomic neighbours, or whose accessibility rises and falls together across many ENCODE cell types. They are not defined from co-expression in the screens.

Null	Role	How the replacement feature is drawn	Exact if	Main weakness
Block permutation	Main	Remove the part of x predicted by C, then shuffle the residuals as intact blocks of genes whose accessibility moves together	Dependence in x stays inside blocks, and blocks are exchangeable	Blocks must be estimated; unequal sizes need matching
Stratified permutation	Sensitivity	Shuffle x among genes within bins of expression difference and copy-number state	x is exchangeable within bins	Ignores that neighbouring and co-regulated genes share accessibility
Decoy features	Sensitivity; H4	Use accessibility differences from cell-line pairs outside the comparison	Decoys share the structure of x but not its tie to these lines	Decoys are finite and related to each other

Decoys are chosen by measured similarity: the correlation between a decoy and x after removing covariates. Decoys with near-zero similarity form the negative control. Decoys across the full range of similarity feed H4.

Effect model. The quantity being estimated is θ in:

$$M_g = \theta\, x_g + f(C_g) + \varepsilon_g$$

θ is the change in response difference per unit of accessibility difference, and f is the covariate adjustment. H1 is the claim that θ is positive.

Confidence interval. For each candidate θ, I subtract θ times x from M and run the test on what is left. The 95% interval is every θ the test does not reject, so it inherits the validity of the test.

Summary measure. Partial R-squared of x, as a share of reproducible variance, is reported as a description, not as the inference. Its interval comes from a bootstrap over knockdowns that recomputes the reliability in every resample. Uncertainty in both the numerator and the reliability is then carried through.

Comparators. Four standard approaches are run on the same data: regression treating genes as independent, unrestricted permutation of x, permutation within expression bins only, and a CAMERA-style correction using inter-gene correlation from control cells.

Multiple testing. The primary family is H1 on the combined feature in the two matched pairs, two tests with Holm correction. H3 is judged by its preset false-positive criterion in Aim 3. Secondary and case-study results are labeled as such and reported in full.

Aim 3: Prove the test works

Two benchmarks with known truth show where each test keeps its 5% false-positive rate and what that costs in power.

Synthetic simulation. I generate response differences and a feature while controlling five settings:

- correlation of responses within gene modules
- how far the feature shares those same modules
- confounding between the feature and expression
- noise that depends on expression and cell count
- the true effect: zero, small, moderate or large

The failure conditions are predicted in advance. Independent-gene tests should inflate only when responses and feature are both structured along the same modules. Expression-bin permutation should fail when the feature has shared structure within bins. Block permutation should hold when blocks are right, so I also test it with deliberately wrong blocks.

Planted signal in real data. Real response differences keep their true noise and correlation, which no simulation fully captures. I add a known effect tied to a decoy feature, then check whether each test recovers it. With no effect added, the same setup measures false positives on real data.

Every test is scored on the same four measures, each with Monte Carlo error bars:

- false-positive rate at the 5% and 1% levels
- power at a matched false-positive rate
- bias of the effect estimate
- coverage of the 95% confidence interval

Each setting is then pushed until the main test fails, and that boundary is reported as the limit of the method. The planted-signal benchmark is also run on each matched pair, where the lines are most alike and power is lowest.

The central figure plots false-positive rate against the strength of shared structure, one line per test. If the naive tests stay near 5% across realistic settings, H3 is wrong and the report says so.

Aim 4: Confirm and extend

The frozen pipeline runs once on the two matched pairs, which carry the main test of chromatin. The rest of this aim puts that result in context.

Matched pairs (primary). K562 and Jurkat are both blood-derived, p53-mutant and grown in suspension. RPE1 and HepG2 are both epithelial, p53 wild-type and adherent, as the TRADE preprint describes them. Within each pair those three factors are held constant, so an accessibility effect cannot stand in for them. H1 is tested here and nowhere else.

What matching costs. Two costs are stated up front:

- **Batch.** Each matched pair joins a line from one study to a line from another. Effects are measured against each line's own controls and the generic response is removed, which limits the problem. The same-line baseline covers two screens from one study, so it does not measure cross-study batch.
- **Power.** Matched lines respond more alike, so there is less difference to explain. The detection limit from Aim 3 is reported for each matched pair.

Mixed pairs (one contrast). The four mixed pairs give a single pooled estimate: the average of their four statistics, with one block permutation applied to all four at once. There chromatin is entangled with p53, tissue and growth mode. A larger effect than in the matched pairs is expected and is not read as chromatin alone. The gap between the two estimates shows how much the confound could carry.

Similarity gradient (H4). For each pair's response difference I estimate the effect of every available accessibility difference: its own, the other five pairs' and the decoys'. Each is placed by its similarity to the pair's own feature, measured as their correlation after removing covariates. Under the effect model the estimate should rise in proportion to similarity and reach zero where similarity is zero. One plot shows effect against similarity for all features. A generic artifact would give a flat line that does not pass through zero.

p53 control. p53 targets should diverge in the four mixed pairs and not in the two matched pairs.

Same line, two states (extension). X-Atlas/Pisces includes genome-scale screens of resting and CD3/CD28-activated Jurkat cells, described in the X-Cell preprint. One line in two states holds genome, p53 status and lab constant, which would remove the cell-line confound entirely. It enters the project only if three things check out:

- the licence permits this use
- ATAC-seq exists for Jurkat cells under a matching stimulation
- screens of about 2.8 million cells each fit the available compute

A 2026 screen of primary CD4+ T cells at rest and after re-stimulation is a possible alternative.

Mechanism case studies (H5). For each transcription factor in the library with ChIP-seq or a reliable motif:

1. Count the factor's binding sites near each gene that are accessible in each line.
2. Take the difference between lines as a feature specific to that knockdown and gene.
3. Compare it with the response difference for that knockdown alone.
4. Repeat with another factor's sites as a swap control. The knocked-down factor's own sites should predict best.

Results are shown factor by factor with no pooled test. K562-HepG2 has the most ChIP-seq, but it is a mixed pair, so these cases illustrate a mechanism and do not establish it.

Interpreting outcomes

Every outcome below yields a reportable result, and the report leads with whichever one the data support.

Outcome	What it means	The report leads with
Positive effect in both matched pairs, rising with similarity	Chromatin context drives divergence apart from p53, tissue and growth mode	Theta, its interval, and the similarity plot
Effect in the mixed pairs only	The apparent chromatin effect is carried by what differs between line classes	The gap between matched and mixed estimates
The generic response accounts for most of the raw difference	Much context dependence is one shared stress and growth response, scaled by line	The generic share for each pair
Naive tests find an effect; the main test does not	Analyses of this kind produce false positives on perturbation data	The inflation result from Aim 3 and the real-data reversal
No effect under any test, with adequate power	Baseline expression already captures what accessibility knows	The smallest effect the study could have detected
Cross-line numbers sit near the same-line baseline	Measured divergence is mostly technical at this depth	The baseline comparison
The two matched pairs disagree	The effect is specific to one pair, or driven by batch	The pair-by-pair comparison and what differs
Signal in distal accessibility only (secondary)	Context dependence sits in enhancers, not promoters	The promoter-versus-distal contrast

A null result is only informative with a stated detection limit. The planted-signal benchmark supplies it: the smallest planted effect the test recovers at 80% power.

Limitations

The study establishes association under a valid test, not cause. Seven limits apply.

- **Direction of cause.** Accessibility and expression programs shape each other, so an association does not show that open chromatin causes a stronger response.
- **Scope.** Four immortalized or cancer lines and essential genes only. Removing the generic response reduces, but may not remove, the stress responses these knockdowns trigger.
- **Cross-study pairs.** Both matched pairs span two studies. No line was screened in both, so that batch effect cannot be measured directly.
- **Mismatched samples.** ATAC-seq comes from different labs and cultures than the screens.
- **Bulk accessibility.** One profile per line averages over cell-cycle and other states.
- **Model dependence.** Exact validity needs the true distribution of the feature. Checking the main null against two others bounds this risk but cannot remove it.
- **Simulation realism.** Synthetic data reflect my modeling choices, which is why the planted-signal benchmark carries more weight.

Independent work

This plan was drafted with AI assistance, and that use has to be logged and disclosed. The research, the code and the written report must be mine.

Before any result goes into a report, I should be able to do each of these unaided:

- prove that the randomization p-value is valid when the replacement features are exchangeable with the real one
- explain why the feature is randomized and the response is held fixed
- explain why subtracting the noise variance removes the bias in squared effects
- explain what the generic-response subtraction removes and why the noise null must pass through it too
- explain why the matched pairs carry the main test and what that choice costs
- write and debug every step of the pipeline
- justify each covariate and say what would change without it
- explain every figure to a statistician and to a biologist

A mentor in computational biology or statistics should check three things: the novelty claim, the proof, and whether the simulations are realistic. Their role is review, and the application has to describe it accurately.

I keep a dated research log from the first download, including every AI prompt used.

Open items

These are unverified and could change the design, so they come before any analysis.

- Count shared knockdowns for each pair from the data files, against the 2,053 TRADE reports
- Confirm the Jurkat and HepG2 screens are downloadable with their control cells
- Count knockdowns shared by the two K562 screens and list how their protocols differ
- Find ENCODE ATAC-seq accessions for Jurkat and HepG2, processed like the K562 data
- Confirm TP53 status and copy-number profiles for all four lines
- X-Atlas/Pisces: read the licence, size the Jurkat download, and search for ATAC-seq of resting and CD3/CD28-activated Jurkat cells
- Count sequence-specific transcription factors in the library that have ChIP-seq in K562 or HepG2
- Measure memory and runtime needed to load one full screen
- Read TRADE, the July 2026 decomposition preprint and ChromaPert in full; search for any prior estimate of this effect
- Contact the 4DN lab about use of the RPE1 ATAC-seq data
- Identify a mentor and agree what their review covers

Sources

- Nadig et al. 2025, TRADE, Nature Genetics and its preprint on PMC, for knockdown counts, cross-line correlations and the description of the four lines
- Perturbation response decomposition, bioRxiv, posted 27 July 2026
- ChromaPert, arXiv 2610.09054, submitted 6 October 2026, confirmed on arXiv
- X-Atlas/Pisces dataset card and the X-Cell preprint
- Genome-scale Perturb-seq in primary CD4+ T cells, Cell 2026
- Multiome Perturb-seq, Cell Systems
- ENCODE K562 ATAC-seq, ENCSR868FGK
- 4DN hTERT-RPE1 ATAC-seq, 4DNES48BA2C8

Replogle et al. 2022, CAMERA, Efron and Tibshirani, Candès et al. and Ahlmann-Eltze et al. are cited from memory and need checking against the papers.